



**National Health Service
Standards 2nd Edition
for Papua New Guinea
Volume Three
Design Standards for
Health Facilities in Papua
New Guinea
2021-2030**

2021

National Department of Health

National Health Services Standards 2nd Edition Volume Three

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Volume 3

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NATIONAL HEALTH SERVICE STANDARDS, 2nd Edition
VOLUME 3 DESIGN STANDARDS FOR HEALTH FACILITIES IN PAPUA NEW GUINEA

Applicable to: All Staff

Last Review Date: 1st QR 2021

Next Review Date: 2030.

Version Control and Document History

Date	Summary of modifications made	Version no
2011	Initial National Health Services Standards Vol 1-2-3	V1
2021	Modified - NHSS 2 nd Edition Vol 1-2-3	V2

Acronyms

Acronym	Definition
GoPNG	Government of Papua New Guinea
PHA	Provincial Health Authority
NDoH	National Department of Health
NGO	Non-Government Organisation
NISIT	National Institute of Standards and Industrial Technology

Glossary of Terms

Term	Definition
Architect	Person who is qualified to designs buildings and supervise their construction
Building By-Laws	Regulations by which local authorities control building construction. No new work or alteration of existing structures may be carried-out unless these regulations are observed.
Civil Engineering Works	Works comprising a structure other than a building and its associated site works, such as a road, foot-pathing, water dam, soil stabilisation, etc.
Contract Documents	Working drawings, specifications and details, which form part of the formal agreement between the Contractor and the Client for the construction of a building/facility.
Contractor	One who agrees by written agreement or contract to supply materials and perform certain types of work for a specified sum of money.
Elevation	A geometrical drawing of the façade of a building/facility.
Engineer	A professionally qualified person, who when associated with building and planning, designs, supervises and manages Civil, Structural, Mechanical, Electrical, and Services components of a Building Construction.

EXECUTIVE SUMMARY

Volume Three of the National Health Services Standards (NHSS) 2021-2030 is to inform all stakeholders involved in the **planning, design, development, commissioning, operation, maintenance and disposal of health facilities infrastructure and medical technology for the public sector in PNG.**

It is expected that the information herein will be of interest and value to Provincial Health Authorities (PHA) health service providers, funding bodies, architectural and engineering consultants and members of the construction industry involved in the design and delivery of new health service infrastructure in PNG.

Most of the facilities from which health services are delivered in PNG at this time are in poor condition and are being systematically replaced through the implementation of the Government's 'Vision 2050' and the National Department of Health's (NDOH) National Health Plan 2011-2020 and its successor NHP 2021-2030.

It is therefore timely that the NDOH disseminates standards for the design and construction of new health infrastructure to ensure that the capital works program consistently delivers facilities which provide value for money, a sustainable asset and a modern clinical environment to support the work of our health professionals.

Compliance with these Design Standards will also:

- Ensure new healthcare facilities provide a clinically safe, culturally appropriate, efficient and comfortable physical environment for clients, visitors and staff members;
- Ensure standardisation of facilities provide familiar environments for staff moving across the sector; and
- Ensure, within reason, the equitable provision of amenities over time across the sector.

AUTHORITY

The Government of PNG has an obligation conferred by Part IV, Clause 18.1 of the National Health Administration Act (NHAA) 1997 to:

- a) Oversee the establishment, maintenance and development of a health care system in the country;
- b) Set policy and fix standards for the improvement of the health of the population;
- c) Provide technical advice and support for the operation of health facilities and the delivery of health services;
- d) Oversee the management of public hospitals in accordance with the Public Hospitals Act, 1994; and
- e) Maintain a national health information system.

To comply with some of these obligations the NDOH is authorised to, among other responsibilities:

- Promulgate Service Quality and Resource Standards to be achieved by all PHA public health care service providers throughout Papua New Guinea;

- Monitor compliance with those standards; and
- License premises for the delivery of health care.

It is therefore appropriate for the Department to articulate 'Standards' for design and construction of health care facilities to ensure that such facilities are 'fit for purpose', suitable for licensing and affordable in both capital and recurrent operating costs and that they, above all, provide a safe and culturally appropriate environment for clients and guardians.

Policy

It is the policy of the NDOH that all new health facilities designed for the public sector after 31st December, 2010 will be designed and constructed to the NHSS design standards to ensure that culturally appropriate, modern, user friendly and cost-effective facilities are provided from which to deliver publicly funded health services.

This will be achieved by providing facilities that allow comfort, privacy and dignity for clients of the health service through:

- The separation of adult male and female patients in general inpatient settings;
- Flexible use of inpatient beds to allow separation of infectious or distressed patients and the provision of private/intermediate care;
- Sufficient discrete and separate ablution facilities for clients, guardians and staff;
- Discrete triage stations for the interview and assessment of clients;
- Acoustically insulated consultation, examination and treatment rooms;
- Discrete access to specific clinics;
- Ensuring that all facilities are fully accessible for disabled clients; and
- Appropriately controlled access to critical care services.

The unique cultural requirements of our people will be provided through the:

- Provision of separate amenities for guardians;
- Provision of culturally appropriate grieving facilities for the families and friends of the deceased;
- Provision of generous outdoor waiting and gathering areas for clients, guardians and extended families; and
- Signage in Tok Pisin, English and Motu where appropriate and the use of international symbols to denote the functions of each component of the health facility.

Wherever appropriate, consultant architects will be encouraged to design facades for health care buildings which reflect the unique building styles of the local people. Health facilities must also provide a clinically safe environment for clients and a safe workplace for staff. This will be achieved by:

- Ensuring that departments, wards and units within redeveloped health facilities are physically located to optimise access to critical supporting services for the delivery of emergency clinical care.

- Ensuring that medical equipment is installed and maintained consistent with the delineated role of the facility.
- Ensuring that adequate facilities are provided to control the spread of infection.
- Installing electrical reticulation, ventilation and engineering systems which ensure patient safety.
- Ensuring that all buildings are constructed in accordance with the Building Code of Papua New Guinea.
- Ensuring that fire prevention, containment and control equipment is installed consistent with relevant Papua New Guinea legislation.
- Providing amenities for staff consistent with relevant industrial legislation.
- Providing duress alarms and physical security barriers in areas where staff may be threatened by aggressive clients or where assets are stored or handled which might attract criminal activity.
- Discrete use of security surveillance infrastructure and technology to assure security for personnel and assets

Generally, any major capital projects for the public health sector will continue to be funded and contractually supervised through the relevant divisions of the NDOH, although, in some instances, Provincial Health Authorities including Public Hospitals or Provincial Administrations may make their own arrangements for delivery of smaller projects. Irrespective of the project delivery arrangements the actual design of healthcare buildings, specification of support systems and development of contract documentation is a task for experienced, professional, consultant architects, engineers and health planners.

It is expected that all PHAs and other health services providers (i.e. Faith-based organisations, Non-Government Organisations and Private Sector Providers) who are constructing or renovating healthcare facilities to provide health services within PNG will comply with these **Design Standards, and will choose to design and construct their facilities consistent with these Design Standards.**

Cost and Accountability

Much of the public sector healthcare infrastructure in PNG at this time has reached or is approaching the end of its useful life for such purposes and it is envisaged in the Government's 'Vision 2050' and the National Health Plan that there will be a large number of significant capital projects planned and delivered within the public health sector during the next 20 years. This will require the commitment of very significant amounts of public funding at a time of competing demands upon Government for other public infrastructure.

It is acknowledged that compliance with these Standards for health projects may result in additional capital costs for some projects but the outcome will be the provision of more appropriate, robust and efficient facilities from which to provide health services. In the longer term it will prove to be a prudent investment.

Most health care facilities are designed with an expected useful life of at least 30 years and most health professionals will have no experience in the delivery of major capital projects or with capital

planning processes during their working life. However, there is a requirement that staff members are consulted during the briefing and throughout the design phases of any major project.

This will be achieved by involving User Groups from the particular PHA Health Service in the briefing of Consultant Architects and Engineers and through regular meetings as design proceeds through its various stages.

In return for this level of involvement in the planning processes the NDOH expects that there will be much more rigour applied by PHA health service staff to establishing their actual requirements than may have been applied in the past.

The cost of providing new facilities and equipment and operating those facilities will be contained by taking a 'business like' approach to the design of new facilities, specification of engineering systems and provision of essential medical equipment

Any proposal for funding to procure major capital projects must be supported by a Business Case identifying the benefits to service delivery and by a Recurrent Cost Study identifying any recurrent budget impacts arising from the implementation of the project.

Capital Planning Process

Capital planning affords the opportunity not only to acquire new or renovated health care facilities but to modernise clinical practice, implement new technologies and services and to rectify any clinical or environmental safety issues which may have been dictated by the existing facilities. It is also an opportunity to improve client access, utility and satisfaction, staffing and resource efficiency and cultural appropriateness.

In order to optimise the opportunities associated with a major capital project a comprehensive review of the Health Service's delineated role, operational policies, clinical service development plans, current service delivery models and the critical functional relationships within and between Departments must be undertaken before any detailed planning commences.

PREPARATION OF BRIEFINGS

Before Consultant Architects, Engineers and Project Managers are engaged for any major capital project the National Department of Health will support the local development of a Design Brief by staff of the particular health service to specify the capacity required, any special design implications arising from the preferred Service Delivery Model to be used in the new or renovated facility and any special functional relationships between and within departments, units or wards. The resulting Design Brief may ultimately serve as the briefing paper for consultants engaged to take the project to fruition.

However, before proceeding to the engagement of consultants for publicly funded health infrastructure the Design Brief will require endorsement by the National Department of Health to ensure that proposed capacity provision is appropriate to the needs of the catchment population, that the proposed facilities are consistent with the Service's delineated role and that the recurrent budget implications have been identified and can be funded. It is expected that Design Briefs for major projects will be reviewed professionally by a Health Planner as part of the compilation of a Project Definition Plan for those particular projects.

Before the completion of a Design Brief for any project the following matters must be considered.

LEVELS OF CARE AND ROLE DELINEATION

The level of care and the role of individual health services is delineated as per the NHSS in order to rationally distribute the scarce human and physical resources available for the delivery of health care.

The delineated role of a particular health service prescribes the complexity of the clinical activity undertaken at that particular service and, therefore, prescribes the level of clinical and general support services that are required and the staffing profile that is required to ensure that clinical services are appropriately supported and safely delivered. Understanding the level of care and delineated role of a particular health service is an essential prerequisite for determining what facilities and equipment are appropriate to be provided for that service.

The NHSS 2011 has a role delineation matrix, NHSS Vol 1 2021-2030 now outlines the six (6) levels of care and it is expected that any future capital projects will provide amenities consistent with the particular health service's level of care and delineated role.

Where a PHA health service has involvement in teaching and development of future health professionals there will be a requirement for additional teaching facilities, the establishment of a PHA Training and Resource Centre, and include tutorial rooms, library etc.

HEALTH SERVICE DEVELOPMENT PLANS AND CLINICAL SERVICES DEVELOPMENT PLAN

Demand for health care changes as new communicable and non-communicable diseases arise and as lifestyles change in response to changing socioeconomic circumstances and community attitudes. Clinical practice and care continue to evolve with the introduction of new technologies, interventions and clinical requisites and as health professionals update their skills to keep abreast of new techniques.

It is therefore imperative before embarking on development of major facilities designed to last up to 30 years that health services development plans and clinical service development plans are developed, and/or revised to take account of emerging health demands within the existing catchment population and foreseen changes that will arise at least over the medium term.

Such changes often necessitate the development of new services/facilities and the introduction of major capital equipment with specific building design requirements.

When finalising these Development Plan, it is important to be realistic about the prospects for attracting the capital budget and the clinical staff required to operate any proposed new services. PHA Development Plans must not be a 'wish list' and particularly where it is likely to drive capital investment.

SERVICE DELIVERY MODEL

Service Delivery Models have generally evolved to suit the existing physical facilities from which services are delivered and the personal preferences of individual clinicians. Over time such service delivery models can constrain innovation, entrench inefficiencies and disenfranchise the 'client'.

The opportunity to renovate or replace health facilities provides an ideal opportunity to refocus the service delivery model to more modern, efficient and client focussed methods of service delivery.

This inevitably means ‘change’ of work practices, facility layouts and functional relationships between departments/wards/units and can be challenging to most staff members, particularly those who have been long serving.

Nevertheless, review of the current service delivery model is another essential prerequisite of the design briefing process for major new health infrastructure.

There are various outpatient services whose scope of practice can overlap unless role clarity is provided from the outset. Ensuring continuity of care for clients, particularly the chronically ill, is paramount and wasteful duplication of effort and facilities must be avoided if limited resources are to be used efficiently.

In this respect, clarification of the role and hours of operation of Emergency Departments, General Outpatient Services (Adult & Children’s Outpatient Departments), Specialist Clinics and Specialised Outpatient Services (STI Clinics, TB Clinics, MCH Clinics etc) is essential before briefing for design commences.

Issues relating to the manner in which inpatients will be aggregated must also be clearly settled before briefing for design commences. Issues such as the cut off age for admission to Paediatric Wards, where and by whom surgical patients (particularly Obstetrics & Gynaecology and Paediatrics) will be cared for, where and by whom oncology patients will be cared for, where private or intermediate patients are to be cared for must all be definitively resolved as such issues are critical to the distribution of beds across the various wards.

The manner in which delivery suites are used and the birthing process is managed will determine the number of delivery beds required and the mix of pre and post natal beds to be provided.

Establishment of facilities for day only surgery and endoscopy and the provision of ‘observation/short stay beds’ within the Emergency Department will have implications for the number of beds required in overnight surgical and medical wards.

The introduction of technologies such as digitised medical imaging distribution and electronic pathology reporting, computerised patient management systems and electronic document storage systems can have profound impacts upon the amount of storage space to be provided for radiographic films, medical and administrative records and the distribution of offices and work stations in clinical settings.

The manner in which clinical and general support services such as retrieval services, sterilising services, biomedical engineering, linen and food services are to be provided will have implications for the design and capital cost of any new facilities.

Issues around the privatisation or outsourcing of some services such as cafeterias, security, patient transport or some engineering functions must also be resolved before briefing for design commences.

OPERATIONAL POLICIES AND GUIDELINES

Existing policies and guidelines which guide the delivery of care, clinical and general support should also be reviewed before briefing for design commences.

Review of existing policy and guidelines relating to issues such as internal referral arrangements, the operation and supervision arrangements for the Intensive Care Unit, bed allocation arrangements, operating theatre access for both elective and emergency procedures, use of operating rooms for

minor procedures, on call arrangements for clinical and allied health staff, provision of meals for 'guardians' is strongly recommended as these can all have significant impacts upon the amount of capital budget required for a project.

Policy and guidelines should be formulated with respect to the amount and types of staff amenities to be provided in particular work sites.

Review of any formal arrangements existing with tertiary education providers is also advisable to ensure that adequate facilities are provided for teaching and staff development activities and offices are available for those academic officers with joint appointments.

Policy and guidelines should be formulated with respect to the availability and regulation of car parking on the campus as this is a provision that is often overlooked as capital projects are developed.

CAPACITY PROVISION

Prior to commencing a major capital project, it is also essential to clarify the extent of the catchment population, the existing demographic and morbidity profiles for that population and the manner in which and by whom health services will be delivered.

Faith Based and other Non-Government Organisations play a significant role in the provision of health services in Papua New Guinea and planning for Public Sector infrastructure must not be undertaken in isolation of those other service providers. Failing to consult other providers about their plans for service delivery may lead to expensive duplication of infrastructure and effort.

Since current activity levels will form the baseline from which any planning for service growth will be calculated it is essential to review the efficiency with which current facilities and resources are utilised. Comparative analysis of admission rates and thresholds, average length of stay of inpatients by diagnosis, effective operating theatre utilisation rates, percentage of surgical procedures performed on the day of admission, day surgery rates, waiting times by triage category in Emergency Departments etc. should be compared with peer health services to ensure that planning is commencing from a sound foundation. Providing facilities in excess of what is required to efficiently deliver health services is wasteful of both capital and recurrent funds.

Currently, the World Health Organisation recommends the provision of 3.3 beds per 1000 of catchment population for countries within the South West Pacific area. However, it is the policy of the NDOH that, in general, a total of 2 inpatient beds per 1000 of the catchment population will be provided for low acuity inpatient care spread across facilities ranging from Community Health Posts, Health Centres to District Hospitals.

An additional 0.5 beds per 1000 of catchment population will be provided for the referral role of PHA Provincial Hospitals and an additional 0.2 beds per 1000 of the catchment population for the referral role of the National Referral Teaching Hospital. In this context, 'catchment population' is defined as *"that group of persons who would attend the health service were they to require treatment from a particular individual or group of practitioners"*.

It is important to recognise that a health service may have different catchment populations for different specialties and that the catchment population will change as neighbouring services expand or collapse the range of services which they are able to offer.

The overall provision of 2.7 beds per thousand of population for PNG rather than the 3.3 beds per thousand recommended by the World Health Organisation is rationalised on the basis that:

- The same level of demand for surgery, particularly orthopaedic implants, cardiac, vascular and cosmetic/restorative procedures is not evident in PNG as in some other South West Pacific nations.
- Demand for treatment of medical conditions such as strokes, diabetes and heart disease is not yet at the levels of developed Nations although there is evidence that these are emerging as future challenges for the Papua New Guinea health system.
- Demand for provision of high-cost recurrent treatments such as renal dialysis is presently moderated by cost, availability and accessibility and is likely to be so for at least the medium term.
- Demand for other high-cost therapies such as chemotherapy and other cancer therapy is limited by cost, availability and accessibility and is likely to be so for at least the medium term.
- Provision of care for terminally ill, aged and disabled persons with chronic illness remain the responsibility of families in PNG society and is generally provided in the family home rather than healthcare institutions
- Whilst the total fertility rate within the PNG population is higher than in most South West Pacific countries, the extent of home birthing is very much higher and the average length of stay in hospitals for uncomplicated births is much less thereby mitigating the number of obstetric beds which might otherwise be required.

In specific instances where a PHA Hospital provides a National Service such as the National Oncology Service at ANGAU Memorial General Hospital additional beds may also be provided based upon actual demand for that particular service.

When calculating the actual bed requirements for individual facilities it must be recognised that PHA Provincial Hospitals and National Referral Teaching Hospital also provide District Hospital Services for their immediate catchment populations in most cases.

The availability of transport infrastructure is also an important determinant of health seeking behaviours and this must be considered when determining where to locate tertiary and secondary health services and outpatient and primary health services.

In those provinces where there is an extensive all weather road network demand for health services is consistently higher at the Provincial General Hospital due to the high rate of self referral in such circumstances.



PROCUREMENT & DESIGN PROCESS

The following processes will be adopted for all PHA publicly funded health projects. Once a Design Brief has been locally developed and endorsed by the NDOH for a Project and the capital funds have been appropriated Officers of the NDOH will assist to facilitate the procurement process for Consultants to progress design, construction, project coordination and supervision.

The design process will generally consist of the following four (4) stages:

- Project Definition Plan
- Master Plan
- Schematic Design
- Final Design

The first stage is the development of a **Project Definition Plan** which defines the scope of the building required to accommodate those services to be provided and is generally derived from a review of the Design Brief by an experienced Health Planner. This Plan will guide design and construction of the new facility.

The second stage is the preparation of a **Master Plan** for the site indicating the location of existing buildings suitable for retention or requiring demolition, the location of existing utilities (water, sewage and electrical main supplies) the proposed location for new buildings and zones for future development. Master Planning is a preliminary and essential element of design for any major project and is undertaken to ensure that opportunities for future expansion or changes to the Health Service's levels of care, delineated role or service profile can be affected without compromising existing structures or services.

Once a Master Plan for the site has been endorsed by the Governing Body and the NDOH the Project will proceed to the Schematic Design stage. **Schematic Design** is preliminary work and provides basic detail of layouts, floor areas and depicts the functional relationships between services in each of the proposed buildings, floors etc. The Schematic Design provides details of gross floor space and therefore allows estimation of likely construction costs at an early stage of the design process. Schematic Design requires direct and significant input from the staff involved in delivery of services from the facility if that facility is to ultimately meet the needs of the 'end user'.

Once Schematic Design has been approved by the Governing Body and the NDOH the Consultants will proceed to Final or Detailed Design. The **Final Design** will ultimately deliver tender documentation including plans and engineering specifications of sufficient detail to enable tenders to be invited for construction. During this stage of design staff involved in the delivery of services from that particular part of the facility must be consulted and must contribute conscientiously to the final design since there will be no further opportunities to ensure that the facility meets their needs and expectations. Post contract variations to the Final Design are generally costly and only authorised in exceptional circumstances.

Once a final design has been formally endorsed by the PHA - Governing Body of the Health Service and the NDOH the project may proceed to tender for construction if sufficient funding is available. Generally, tendering for construction will be managed by the relevant Provincial Procurement

Committees or the National Procurement Commission Board dependent upon the projected cost of construction.

BUILDING STANDARDS

LEGISLATION AND APPROVALS

These Design Standards have been largely derived from the Australasian Health Facility Guidelines and, as such, they comply with Australian and/or New Zealand legislation with respect to disabled access, fire safety, occupational safety and engineering Standards.

Generally, Papua New Guinea has endorsed the same Standards but where separate standards exist in Papua New Guinea compliance with the local standard is required.

All building works must comply with the '*Building Code of Papua New Guinea*' and local planning laws must be observed.

The consent of the relevant 'Provincial Building Board' must be arranged before construction commences and periodic inspections must be conducted by Officers of the Board during the construction phase. Generally, the Building Board will also assess plans for compliance with local fire safety regulations.

Connection to mains supplies of electricity, water and sewage systems requires formal approval from Papua New Guinea Power and the Papua New Guinea Water Board or relevant Town Authority.

Where it is proposed to discharge treated effluent from the site the prior approval of the Department of Environment and Conservation is required. In some cases, mains may require renewal or upgrading and this work can delay projects if it is not harmonised with the construction program. Similarly, arrangements must be made early in the process with Telikom PNG if upgraded or additional land line communication capacity is required.

Where it is proposed to install a high temperature incinerator for the destruction of clinical waste the prior approval of the Department of Environment and Conservation is required.

ARCHITECTURAL FEATURES

Consultant Architects will be encouraged to explore local customary building styles and to incorporate, wherever feasible, external features which reflect the local culture and traditions. This may include traditional roof lines, facades at major public entry points and traditional ornamentation of supporting columns provided that additional capital costs are modest and maintenance can be cost effectively managed.

DESIGN AND CONSTRUCTION FOR RURAL AND REMOTE AREAS

Constructing permanent buildings in the rural and remote areas of Papua New Guinea can be logistically challenging and costly. The logistical issues associated with delivery of building materials to site can be immense with many locations only accessible by light aircraft, helicopter or on foot.

In most rural and remote areas there is no access to mains electricity, reticulated water or to building materials such as ready mixed concrete, prefabricated components such as windows and doors, custom made joinery and cabinetry.

Furthermore, experienced and competent Tradespersons are difficult to source and often reluctant to work in such areas.

Many parts of PNG are also subject to heavy termite infestation and there is generally no fire fighting capacity available in the rural areas. These factors contribute significantly to extended construction periods and costs and must be considered when designing buildings and specifying building materials.

Wherever practical and economically feasible, buildings should be constructed of materials which is cost effective, durable, long lasting, provide termite resistance and have a low flammability rating. Provision should be made for the capture and reticulation of rainwater and for on site generation of electricity.

Design should include electricity and water saving methods and technologies. Wherever possible all plumbing and drainage systems should be located on external walls to facilitate maintenance. Solar power systems can be successfully used for lighting and operation of small appliances and two-way radios in the rural areas and solar generated hot water is a preferable alternative to gas or electrically powered hot water systems.

Optimal use should be made of natural ventilation and lighting through the use of louvered windows, skylights and ceiling ventilation systems.

Adequate consideration be given to accessibility, basic services, security concerns, competency of local communities, and possibility to attract energetic health workforce and most importantly, the low cost for operation and its sustainability.

Basic systems for operations, as compared to automated systems/solutions with minimal training and low maintenance cost using local skill should be considered with the possibility for expansion/improvement/urbanization upon the increase of local knowledge, availability of basic services on power/water/telecommunications, etc to be factored in design.



ENGINEERING STANDARDS

Engineering Services Design Guidelines of the Australasian Health Facilities Guidelines and International Health Facilities Guidelines complemented by the Australia/New Zealand Standards, International Standards Organization, European Standards, PNG Standards through the PNG National Institute of Standards and Industrial Technology (NISIT) and relevant industry standards for:

- civil engineering
- structural
- hydraulic
- mechanical
- electrical (including heating, ventilation, and air conditioning)
- information and communication
- fire, medical equipment
- medical gases, medical suction,
- transportation and
- other relevant international standards are streamlined to the needs of the client and used for the design of health facilities.

The general engineering services guidelines are provided as follows:

Engineering Standards as documented below is dependent on the level of the facility, location (remote, rural, urban), access, and viability of mains power and water.

CIVIL ENGINEERING SERVICES

The health facility site is subject to technical investigation to mitigate the risks of:

- Flooding,
- Chemical, asbestos and other hazardous contamination,
- Unexploded Ordinances (UXOs),
- Effects of natural disasters,
- Adverse environmental and geotechnical engineering conditions and
- Unsuitable soil foundation for buildings.

Roads, Paved Yards, Car Parks and Pathways (relevant to the location and site) provide safe access to car park, entrance, service delivery point, maintenance delivery point, emergency service delivery point and emergency evacuation assembly point.

Pathways are linked with any adjacent public transport stops. Roads and pathways have grades, tactile indicators and other design characteristics to comply with disabled access requirements. Roads do not double as pedestrian access ways.

The health facility site is provided with:

- Storm water drainage to prevent flooding of buildings and pooling of storm water on any paved area or recreational space,
- Sub-soil drainage to prevent rising damp or flooding of any basement spaces and
- Design of facilities to cope with 100-year worst storm conditions and to place buildings above theoretical safe maximum flood level.

STRUCTURAL ENGINEERING SERVICES

The Client/Provincial Health Authority determines the planned life of the facility as this impact on the imposed loadings to be adopted in the design and on corrosion protection measures to be specified for the project.

The loadings to be applied to the structural design is not less than the minimum required by the PNG Building Code/Building Code of Australia, relevant Australian and International Standards but may be greater if the building has to serve a post disaster function or where the planned life exceeds 15 years.

The building structure has adequate foundations suited to soil conditions of the site that provide resistance to load to achieve settlements within the limits nominated in the Australian Standards for the defined service life and possesses sufficient redundancies and adequate ductility to prevent progressive collapse.

Structural drawings and specifications at least define:

- Design codes used in the design,
- Design live loading including service loads,
- Wind loading parameters used for determining loads; (region, terrain, category, shielding multiplier, topographic multiplier, importance multiplier),
- Earthquake parameters used for determining loads,
- Any imposed construction/erection loadings, e.g. earth moving equipment,
- Any load limitations applying to use of areas,
- Foundation design parameters,
- Required concrete strength, slump and cover to reinforcement,
- Welding categories,
- Corrosion protection treatment
- Any other details necessary to define the action and performance of the structure,
- Demolition requirements for any demolition associated with the project and
- Erecting sequence requirements for structures requiring a specific sequence.

HYDRAULIC ENGINEERING SERVICES

The Client/Provincial Health Authority defines the extend of hydraulic services to be provided (relevant to the location, site and access to mains power and water) and the performance required from them, which are not less than required by the Building Code of Australia (BCA), PNG Building Code, other statutory regulations and relevant international standards and best practices. The extent of hydraulic services may include but not limited to:

- Cold water service,
- Hot water service,

- Warm water service,
- Water filtration and conditioning equipment,
- Water storage tanks,
- Gardens and grounds landscaping,
- Bore water supplies,
- Sanitary drainage service,
- Process waste water discharge conditioning facilities,
- Sanitary fittings and fixtures,
- Roof plumbing,
- Storm water drainage,
- Sewerage treatment facilities and
- Natural or liquefied petroleum gas services.

ELECTRICAL ENGINEERING SERVICES

All levels of health services must have a source / or combination of power sources, i.e. solar, wind, hydro, mains power.

The Client / Provincial Health Authority defines the extent of electrical services to be provided and the performance required from them which is not less than required by the clinical function, Australia and New Zealand Standards, statutory regulations and other relevant international and industry standards.

The extent also is dependent on the level of service, and the types of basic / complex medical equipment /plant to be provided.

Electrical services may include:

- Provision of normal, vital (30sec), instantaneous (1sec), and uninterruptible (No Break) electricity supplies,
- Switchgear and circuit protection to safely operate and control supplies,
- Distribution arrangements to supply electricity to each end use,
- Equipment to transform and condition voltages from supply voltage to end use voltage and within voltage frequency tolerances,
- Equipment to use the electricity for lighting, heating and motive power and
- Where an electricity supply is denoted as being on an “Essential Supply” then this shall be arranged as a vital (30 sec) supply.

Electrical systems comply with:

- AS/NZS 3000 Wiring Rules
- AS/NZS 3003 Electrical Installations – Patient treatment areas of health facilities
- AS/NZS 3009 Electrical Installations – Emergency power supplies in health facilities.

INFORMATION COMMUNICATION TECHNOLOGY ENGINEERING SERVICES

The Client / Provincial Health Authority define the scope of information communication and technology services to be provided and the performance required from them.

The Health Facility Design Standard room layout and room data sheet compliment choice of such services to be provided.

Data communication and network linking information and computing services are defined by the client.

Communication services required may include but not limited to:

- Health radio communication system,
- Assistance call system,
- Emergency Call system,
- Patient Nurse Call system,
- Building Services Monitoring,
- Data Communications,
- Public Address system
- Duress alarm systems and
- Telephone communication system.

SECURITY

The Client/ Provincial Health Authority defines the extend of security services to be provided and the performance required of them, which are not less than as required by statutory regulations and the extent of services needed for:

- Asset tracking systems,
- Crime prevention through environmental design,
- Door intercommunication systems,
- Security lighting (external and internal),
- Security fences (Perimeter site fence and around any areas to be isolated),
- Security screens (windows, doors and around any areas to be isolated),
- Duress alarm systems,
- Secure parking and
- Security response resources and procedures.

MECHANICAL ENGINEERING SERVICES

The Client / Health Authority specifies the extent of mechanical services to be provided and the performance required from them which are not less than required for the clinical outcome, the Building Code of Australia, PNG Building Code, other statutory regulations and relevant international standards and best practices.

Mechanical services may include but not limited to:

- Air cooling and heating services,
- Building automation control systems,
- Cool and freezer rooms,

- Energy management systems,
- Fume and dust extraction systems,
- Smoke control systems,
- Steam systems,
- Ventilation services,
- Water Sprinkler (fire) Systems, and
- Water treatment systems.

ACOUSTIC

Noise levels of emergency power generators and other equipment/machineries used complies with AS/NZS 2107 and PNG Building Code and other relevant and related standards with respect to noise pollution in a health care setting.

Acoustic isolation between spaces prevents the noise level in one space transmitting to an adjacent space and exceeding the allowable level in that space in order to maintain privacy and patient confidentiality.

FIRE ENGINEERING SERVICES

The client/Provincial health authority defines the extent of fire services to be provided and the performance required from them which is not less than required by the PNG Building Code, other statutory regulations, Fire and Emergency Services, the Health Facility Design Standards and site-specific Risk Management Plan.

Extent of fire services provided comply with the requirements of the PNG Building Code and the Client's site-specific risk management plan and may include but not limited to:

- Provision of materials and methods of construction complying with codes and regulations
- Compartmentalization of the building(s) into fire and smoke control compartments
- Provision of complying fire egress arrangements
- Provision of fire and smoke alarms
- Storage arrangements for fire fighting water
- Fire fighting water pressure boosting arrangements
- Provision of smoke clearing ventilation
- Smoke mode controls for ventilation plant
- Provision of emergency warning and information equipment
- Provision of hose reel and hydrant fire extinguishing equipment
- Provision of automatic fire extinguishing (sprinkler) system
- Provision of portable fire extinguishers and fire blankets
- Provision of equipment to aid transportation of disabled persons
- Provision of escape diagrams.

MEDICAL GASES

Dependent on the level of the facility, location (remote, rural, urban), access, complexity of clinical services and medical equipment supplied.

The Client / PHA specifies the extent of medical gas services to be provided and the performance required from them, which is not less than required for the clinical functions, relevant international standards and by statutory regulations.

Medical gas services may include but not limited to:

- Oxygen storage and reticulation
- Nitrous Oxide storage and reticulation
- Medical breathing air storage and reticulation
- Medical breathing air compression and conditioning
- Medical suction pumping storage and reticulation
- Nitrogen systems
- Carbon dioxide systems
- Industrial and industrial compressed air systems
- Laboratory special gas supplies

Medical gases services comply with AS 2896 Medical gas systems – Installation and testing of non-flammable medical gas and AS 2568 Medical gases – Purity of compressed medical breathing air. Installation and commissioning are by specialist organizations certified to AS/NZS ISO 9002 by a third-party certifying body.

MEDICAL SUCTION

The Client / Health Authority specifies the extent of medical suction services to be provided and the performance required from them, which is not less than required for the clinical functions, relevant international standards and by statutory regulations.

Medical suction services may include but not limited to:

- Medical suction plant and reticulated wall suction outlets
- Active Scavenging wall or pendent outlets
- Portable suction machines.



BIOMEDICAL ENGINEERING STANDARDS

Biomedical Engineering (BME) or Medical Engineering is the application of Engineering Principles and design concepts to medicine and biology for healthcare purposes in such case the clinical and public health care services as delineated by the NHSS.

The **Biomedical Engineering Services and Standards Functions** to be carried out will include:

- Medical Device Regulations,
- Biomedical Equipment Assets Information system,
- Planning for Replacement of Biomedical Equipment,
- Acquisition and Replacement of Medical Equipment,
- Equipment Maintenance Planning and Management,
- Medical Device/Equipment Assets Disposal,
- Biomedical Engineering staff training and development,
- Research into new medical technology, trial of new medical technology,
- Equipment audit, medical device vigilance and adverse effects reporting,
- procurement of spare parts and services,
- medical equipment project management and contract administration,
- Medical Technology ICT/Tele-health.

The regulations of medical equipment/device in the country will be in compliant with International Medical Equipment/Device regulations of ISO 13485, Therapeutic Goods Administration (TGA), Food and Drug Administration (FDA), European Commission (CE), Good Manufacturing Practices (GMP), relevant European Standards and other recognized international standards.

MEDICAL EQUIPMENT, CLINICAL INSTRUMENTS AND STATIC PLANT

Medical Equipment, Static Plant Equipment, Medical Furniture & Fittings is determined by the Client / PHA is based on the health services role delineated to a facility or function and the performance required for each item is be less than as required by the prescribed health service, relevant international and industry standards and statutory regulations.

The necessary steps for the scope required for a fully functional health facility for Medical Equipment, Clinical Instruments and Static Plant is outlined below:

- I. Supply Health Facility Medical Equipment, Clinical Instruments and Static Plant to the health facility site, with equipment specifications shall at least define:
 - Equipment type description
 - Functional specifications
 - Specific technical specification and data
 - Accessories and spare parts
 - The design codes to be complied with
 - The site conditions to apply
 - The performance and quality of the service to be delivered.

- II. Ensure freight, handling, insurance, customs duties, storage and delivery of the Health Facility Medical Equipment, Clinical Instruments and Static Plant,
- III. On-site installation of the Health Facility Medical Equipment, Clinical Instruments and Static Plant,
- IV. Test-run, calibrated, and certified for use, Health Facility Medical Equipment, Clinical Instruments and Static Plant (irrespective of the level of service be that Level 1 – Level 7),
- V. Ensure commissioning of the Health Facility Medical Equipment, Clinical Instruments and Static Plant Hospital Plant for use,
- VI. Conduct medical equipment care and operation training for Health Facility Medical Equipment, Clinical Instruments and Static Plant users, in collaboration with the Provincial and NDoH MSD HFB biomedical technicians and
- VII. Provide onsite technical training to staff on the use of and the maintenance of Health Facility Medical Equipment, Clinical Instruments and Static Plant.

TRANSPORTATION

The Client brief defines the extent of transportation services to be provided and the performance required from them, which is not less than as required by statutory regulations and the health facility design standards.

The extent of transportation services shall include (but not be limited to):

- Ramps
- Lifts
- Hoists
- Carts
- Trolleys
- Wheel Chairs
- Mobile storage containers
- Vehicles
- Ambulances
- Helicopter emergency landing pads/sites



INTERPRETATION OF THE STANDARDS

ROOM DATA SHEETS

The Standards for particular rooms within health facilities are specified in the Room Data Sheets' and 'Room Layout Sheets'.

The '*Room data sheet*' firstly provides detail of the floor area, hours of operation, occupancy and a description of the purpose of each room and any special requirements.

The '*Room Fabric and Outline Specifications*' section provides details of the recommended building materials and finishes to be used in order to provide maximum durability and reduce maintenance commitments.

The '*Fixtures, Equipment and associated Services*' section provides detail of any permanent fixture to be provided in that particular room and the 'services' required to be available within the room to support those fixtures.

The '*Services Schedule*' includes information about access to electricity, data networks, cold water supply, hot water supply, warm (temperature controlled) water supply, taps, drains and reticulated medical gases.

In many health facilities in the rural and remote areas, some of these services will not be technically sustainable and should not be specified.

Consultants are expected to use discretion when specifying services such as air conditioning, negative and positive ventilation systems, body protected electrical systems, medical gas reticulation, temperature-controlled water systems and data networks.

Television reception should be restricted to single bed rooms and teaching facilities in those facilities where mains supplied electricity is available.

Consultation with the 'end users' of these rooms with regard to the 'services' required is very important.

ROOM LAYOUT SHEETS

The room layout sheets provide a template for design of the particular room and indicate fixtures and furniture placement.

This information is presented in a range of 'elevations'.

Room layout sheets for highly specialised clinical environments such as Radiation Therapy Bunkers, Coronary Care Units and Cardiac Catheterisation Laboratories have not been included at this time. When required the National Department of Health will provide specific standards for such facilities.

Architects from the National Department of Health are developing Room Data and Room Layout sheets for 6-Bed and 8-Bed Rooms and for Delivery Suites with multiple birthing beds.

Standard data and layout sheets are under development for Aid Posts, Community Health Posts, Maternal and Child Health Clinics, and STI Clinics having regard to existing standards for such

facilities. They will be made available as supplements to these Design Standards as they are completed.

E-copies of these Room Data Sheets and Room Layout Sheets can be viewed at www.healthdesign.com.au and an editable version is available for a modest fee.

PERIODIC REVIEWS

The Room Data Sheets and Room Layout Sheets are those approved by the National Department of Health, as at 31st December, 2010.

From time-to-time Officers of the Medical Standards Branch of the NDOH will review these to ensure that they continue to represent current '*good practice*' with respect to suitability for clinical practice, building safety and utility and occupant safety. When necessary, they will be amended, reprinted and distributed to all stakeholders.

From time-to-time additional standard Layout Sheets and Data Sheets will be added as preferred standard designs for 'Community Health Posts', 'STI Clinics' and hospital type facilities such as kitchens, laundries etc are developed.



HEALTH FACILITY REGISTRATION

Each health facility in PNG must be registered by the National Health Board.

NOTE: If a health facility is constructed and/or renovated, and remains classified as “Unregistered” it is unable to be staffed with the appropriate health workers/clinicians.

Approval to establish a new or upgraded health facility is required from the body responsible for health services in the Province¹ and the following steps must be ensured:

- a. Establishing new facilities (or upgrade of existing health facilities) is identified within the PHA Provincial Health Services Plan²
- b. Seek approval to proceed with building a new facility (or upgrade to a higher-level facility) and the issuing of a Provisional Facility Number by the NDoH and it is in accordance with the NDoH Health Facility Registration Guidelines which includes:
 - a. A compliance audit of the services and design prior to commencing building stages
 - b. Approval/noting of the relevant NDoH branches overseeing: Medical supplies procurement; Church Operational Grants and Church Salary Grants (if a CHS); workforce and training requirements. The Monitoring and Research Branch is to allocate facility numbers (including provisional numbers)
 - c. A permanent registration number for each new and upgraded health facilities is subject to a compliance audit (i.e. commissioning and post occupancy evaluation) and relevant recurrent funding and budgets
 - d. All new health facilities require final approval by the National Health Board and the Governance Committees of the NDOH will advise all Partners of the decision of the National Health Board
 - e. **The Medical Registration Board is responsible for the licensing of private hospitals in PNG.**

When the Health Facility is fully functional and adequately staffed the next steps are to be finalized:

- Ensure that the Health Facility is fully registered with the NDoH, and if it is a private facility it is licensed with the Medical Registration Board in accordance with the NDoH Health Facility Registration Guideline
- Ensure that there is a functional Health Facility Committee to support the health service and staff
- Ensure that there is regular formal supervision from the Officer in Charge at the supervising Health Centre, or a Program Officer

¹ PHA Provinces – PHA Board through the CEO; Non-PHA Provinces – PEC via the PMT and the Provincial Health Adviser

² Service planning considers health needs, accessibility and community structures and needs. Service Plan reflect the Provincial Development Plan which also reflects relevant Government of PNG plans and requirements.

CONCLUSION

Until this time there have been no officially endorsed 'Design Standards' for health infrastructure in Papua New Guinea. The small number of significant capital developments within the Public Sector delivered since Independence was achieved in 1975 have been designed and delivered by Donor Partners and Non-Government Organisations without the guidance of local 'Standards'.

As a consequence, they have not always met the cultural needs of our clients or the clinical preferences of our clinicians.

These current Design Standards have now been developed through consultation with many of the senior clinicians engaged in the PNG Health Sector's Public Hospitals and Rural Health Services, and whilst these standards have been largely drawn from existing international standards, they have been modified where appropriate to reflect the needs of PNG clients and clinicians.

As the PNG Health Sector continues its major capital works program for health infrastructure, it is timely for the '*Design Standards for Health Infrastructure in Papua New Guinea*' to be published for implementation by all PHAs and other health services providers, to ensure that all future health infrastructure is purpose-built and meets the specific needs of the next generation.

ANNEXES

Annex 1: General Application of 'Standard Components'

STANDARD COMPONENTS Room Layout Sheets	RLS ROOM CODE	AREA SQ. M	NATIONAL REF. HOSP	PROVINCIAL GEN. HOSP	DISTRICT HOSPITAL	HEALTH CENTRE	URBAN CLINIC	COMMUNITY HEALTH POST	HEALTH POST
1 Bed Room –Outboard Ensuite	1 BR-ST-B	15.0	√	√					
1 Bed Room – Isolation-Negative	1 BR-IS-N	15.0	√	√					
1 Bed Room – Isolation -Positive	1 BR-IS-P	15.0	√						
1 Bed Room – Mental Health	1 BR-MH	14.0	√						
2 Bed Room	2 BR-ST	25.0	√	√	√	√		√	
4 Bed Room	4 BR-ST	42.0	√	√	√	√		√	
6 Bed Room	6 BR-ST	44.0	√	√	√	√		√	
Airlock – Entry – 6m2(Op Room)	AIRLE -6	6.0	√	√	√				
Anaesthetic Induction Room	ANIN	15.0	√	√	√				
Assembly/Prep'n-Pharmacy 20m2	ASPR-20	20.0	√	√					
Audiology Testing Room	AUDIO	14.0	√						
Bathroom	Bath	15.0	√	√					
Bay - Beverage, Enclosed	BBEV-ENC	5.0	√	√	√	√	√		
Bay - Beverage, Open Plan	BBEV-OP	4.0	√	√	√	√			
Bay - Handwashing, PPE	BHWS-PPE-1	0.78	√	√	√	√	√		
Bay - Handwashing, PPE -2	BHWS-PPE-2	0.55	√	√	√	√	√	√	√
Bay - Handwashing, Type A	BHWS-A	0.55	√	√	√	√	√	√	√
Bay - Handwashing, Type B	BHWS-B	0.55	√	√	√	√	√	√	√
Bay - Handwashing, Type C	BHWS-C	0.55	√	√	√	√	√	√	√
Bay - Resuscitation Trolley	BRES	1.52	√	√	√	√	√		
Bay – Storage, 3m2	BS-3	3.0	√	√	√	√	√	√	
Birthing Room - LDR	BIRM	28.0	√	√	√	√		√	
Change - Staff (Male/Female) 10	CHST - 10	10.0	√	√	√				
Change Cubicle - Accessible	CHPT - D	4.0	√	√	√				
Change Cubicle - Patient	CHPT	2.0	√	√	√				
Clean Utility,12m2	CLUR -12	12.0	√	√	√	√	√	√	

STANDARD COMPONENTS Room Layout Sheets	RLS ROOM CODE	AREA SQ. M	NATIONAL REF. HOSP	PROVINCIAL GEN. HOSP	DISTRICT HOSPITAL	HEALTH CENTRE	URBAN CLINIC	COMMUNITY HEALTH POST	HEALTH POST
Clean-up Room, 7m2	CLUP - 7	7.0	√	√	√	√			
Clean-up Room, Pathology	CLUP - P	12.0	√	√√					
Cleaner's Room - 5m2		5.0	√	√	√	√	√	√	
Cleaner's Room - 10m2	CLRM - 10	10.0	√	√	√	√	√	√	
Communications Room	COMM	12.0	√	√					
Consult Room	CONS	12.0	√	√	√	√	√	√	√
Consult Room - ENT-OPT	CONS-ENT-OP	12.0	√	√	√				
CT Scanning – Control Room	CTCR	6.0	√						
CT Scanning – Procedure Room	CTPR	45.0	√						
Dark Room	DARK	6.0		√	√	√	√		
Daylight Processing	DPRO	16.0	√						
*Dental Surgery,14m2	DENSR -14	14.0	√	√	√	√	√		
*Dental Workroom	DENW	12.0	√	√					
Dirty Utility - Sub	DTUR-S	8.0	√	√	√	√	√	√	
Dirty Utility, 10m2	DTUR -10	10.0	√	√	√	√	√	√	
Disposal Room, 8m2	DISP - 8	8.0	√	√	√	√	√	√	
Ensuite - Standard	ENS-ST	5.0	√	√	√	√		√	
Ensuite - Shared	ENS-SH-A	6.0	√	√	√	√		√	
Ensuite - Birthing Room	ENS-BR-A	10.0	√	√	√	√			
Ensuite –Birthing Room – Altern.	ENS-BR-B	10.0	√	√	√	√			
Equipment Clean-up, 8m2	ECL-8	8.0	√	√					
*Film Processing, View & Report	FPVR	16.0	√						
*General X-Ray	GENXR	30.0	√	√	√	√			
Goods Receipt - Pharmacy	GRE	5.0	√	√					
Gymnasium – Physio 45m2	GYAH-45	45.0	√	√	√				
Interview Room –Large/Family	INTF	12.0	√	√	√				
Mammography Room	MAMMO	16.0	√	√					
Meeting Room – Small, 9m2	MEET-9	9.0	√	√	√	√	√		
Meeting Room – Med/Large,15m	MEET-L-15	15.0	√	√	√				
Meeting Room – Med/Large,20m	MEET-L-20	20.0	√	√					

STANDARD COMPONENTS Room Layout Sheets	RLS ROOM CODE	AREA SQ. M	NATIONAL REF. HOSP	PROVINCIAL GEN. HOSP	DISTRICT HOSPITAL	HEALTH CENTRE	URBAN CLINIC	COMMUNITY HEALTH POST	HEALTH POST
Meeting Room – Med/Large,30m	MEET-L-30	30.0	√	√					
Mortuary – Autopsy Room	MOR-AU	30.0	√	√					
Mortuary – Clean-up Room	MOR-CU	6.0	√	√					
Mortuary - Cool Store	MOR-CS	25.0	√	√	√				
Mortuary Concept Plan	MOR-PL	-----							
Neonatal Bay – General Care	NBGC	5.0	√	√	√				
*Neonatal Bay - Special Care	NBSC	10.0	√	√					
Office - 2 Person Shared	OFF - 2P	12.0	√	√	√				
Office - 3 Person Shared	OFF - 3P	15.0	√	√	√				
Office - 4 Person Shared	OFF - 4P	20.0	√	√	√				
Office – Clinical Handover	OFF - CLN	15.0	√	√	√	√			
Office - Single Person, 9m2	OFF - S9	9.0	√	√	√	√	√	√	
Office - Single Person,12m2	OFF – S12	12.0	√	√	√				
Operating Room - Standard	ORMS	36.0	√	√	√				
Operating Room - General	ORGN	42.0	√	√					
Overnight Stay - Bedroom	OVBR	10.0	√	√	√				
Overnight Stay - Ensuite	OVES	4.0	√	√	√				
Patient Bay - Acute Treatment	PBTR-A10	10.0	√	√	√				
Patient Bay - Non Acute Treat	PBTR - NA	10.0	√	√	√	√	√		
Patient Bay – Holding, 6m2	PBTR-H-6	6.0	√	√	√				
Patient Bay – Recovery, Stage 1	PBTR-RS1	9.0	√	√	√				
Patient Bay - Resuscitation	PBTR-R	25.0	√	√					
Pharmacy Counter	PHA-CO	9.0	√	√	√				
Plaster Room	PLST	14.0	√	√					
Play Area – Paediatric,15m2	PLAP-15	15.0	√	√					
Preparation/Set-up Room (Image)	PRES-S	9.0	√	√					
Procedure Room 16m2	PROC-16	16.0	√	√	√	√	√	√	
Procedure Room 20m2	PROC-20	20.0	√						
Reception/Clerical 10m2	RECL-10	10.0	√	√	√				
Reception/Clerical 12m2	RECL-12	12.0	√	√					

STANDARD COMPONENTS Room Layout Sheets	RLS ROOM CODE	AREA SQ. M	NATIONAL REF. HOSP	PROVINCIAL GEN. HOSP	DISTRICT HOSPITAL	HEALTH CENTRE	URBAN CLINIC	COMMUNITY HEALTH POST	HEALTH POST
Reception (Emergency) 20M2	REC-E	20.0	√						
Screening Room (Fluoroscopy)	SCRN	36.0	√	√					
Scrub Up/Gowning 6m2	SCRB-6	6.0	√	√	√				
Scrub Up/Gowning Shared	SCRBS	10.0	√	√					
Seclusion Room	SECL	14.0	√	√					
Security Room 10m2	SECR-10	10.0	√	√					
Set-up Room 8m2	SETUP-8	8.0	√	√					
Shower - Decontamination	SHDEC	8.0	√	√					
Shower - Patient	SHPT	4.0	√	√	√	√	√	√	
Shower - Staff	SHST	3.0	√	√	√	√			
Specimen Collection Bay, 9m2	SPECC	9.0	√	√					
Specimen Reception/Sort/Prep'n	SPREC	20.0	√						
Staff Room, 15m2	SRM-15	15.0	√	√	√	√	√		
Staff Room, 25m2	SRM-25	25.0	√	√					
Staff Station, 10m2	SSTN-10	10.0	√	√	√	√		√	
Staff Station, 14m2	SSTN-14	14.0	√	√					
Store – Accountable Drugs	STAD	5.0	√	√	√	√	√	√	
Store – Bulk, 20m2	STBK-20	20.0	√	√	√	√			
Store – Bulk, 40m2	STBK-40	40.0	√	√					
Store – Cleaner's	STCL	12.0	√	√	√	√			
Store – Disaster Equipment	STDE	8.0	√	√					
Store – Drugs,5m2	STDR-5	5.0				√	√	√	√
Store – Drugs,10m2	STDR-10	10.0	√	√	√				
Store – Equipment, 14m2	STEQ-14	14.0	√	√	√	√			
Store – Files, 8m2	STFS-8	8.0	√	√	√	√	√		
Store – Files, 20m2	STFS-20	20.0	√	√	√				
Store – General,9m2	STGN-9	9.0	√	√	√	√	√	√	
Store – Sterile Stock,12m2	STSS-12	12.0	√	√					
Store – Sterile Stock,20m2 (CSSU)	STSS-20	20.0	√	√					
Stress Testing	STRT	12.0	√	√					

STANDARD COMPONENTS Room Layout Sheets	RLS ROOM CODE	AREA SQ. M	NATIONAL REF. HOSP	PROVINCIAL GEN. HOSP	DISTRICT HOSPITAL	HEALTH CENTRE	URBAN CLINIC	COMMUNITY HEALTH POST	HEALTH POST
Toilet - Accessible	WCAC	5.0	√	√	√	√			
Toilet – Public, 3m2	WCPU-3	3.0	√	√	√	√	√	√	
Toilet - Staff	WCST	3.0	√	√	√	√	√	√	
Treatment Bay, Chemotherapy	TRMT-CHE	9.0V	√	√					

* Subject to further advice from independent experts practising in these various fields

Annex 2: Room Data Sheets

As per the NHSS 2011 – 2020 Room Data Sheets

Annex 3: Room Layout Sheets

As per the NHSS 2011 – 2020 Room Layout Sheets